

Chapter 4

3 (5 points)

Describe the types of information that can be obtained from the vital registration system of the United States.

The vital registration system of the US gets information on deaths, births, marriages, divorces and fetal deaths.

4 (10 points)

What is the purpose of surveillance systems? Describe the components of a surveillance system. Discuss the limitations of data collected from surveillance systems.

Figure 4.6 in the textbook gives a perfect diagram for the components of a surveillance system. It starts with the actual occurrence of a health event and of course confirmation of the event. Next there are a wide variety of reporters such as healthcare providers, schools, hospitals etc, that report these cases to a wide variety of data recipients. The data recipients are either county, state or federal health department that are then in charge of the data management and can then give feedback to the reporting sources as well as spread information to the public to take action.

Overall the purpose of these systems is to identify trends early and mobilize rapid response mitigating the morbidity and mortality of diseases.

However a major flaw of these systems mainly come from either incorrect reports (ie providers failing to fill out the forms for all cases) or simply patients not seeking medical attention. The latter can be from a variety of reasons such as lack of access to healthcare, patient being asymptomatic, or even choosing not to go due to monetary restrictions. These reasons can cause a slight under reporting of the actual amount of cases compared to the reported amount.

9 (10 points)

Define the term 'big data'. Using your own ideas, state which data sources might be included in big data. What do you believe are some of the advantages and disadvantages of using big data for epidemiologic research?

First off big data is defined by the 3 v's:

1. Volume: large quantity of data

2. Velocity: high frequency of new data points
3. Variety: different types of data (structured, unstructured)

Essentially, big data refers to large quantities of data that are rapidly updating in a wide variety of data formats. Examples of data sources could be social media posts, traffic data, weather data, etc.

For epidemiologic research there are pros and cons to using big data. For starters, the quality of the data is highly variable based on the reputability of the sources as well as the overall completeness or representativeness of the data (is the data an accurate representation of the studied population or just a subset). Additionally, certain datasets are only it is important to use the data for the right use cases. Additionally, there are different ethical concerns when it comes to lack of informed consent and that correlation doesn't prove causation. Also, due to the nature of large data it may take some time to transform the data into a more usable data type (example could be a JSON format), because often times big data is unstructured. However despite all of these downsides there are a lot of benefits of big data such as having a wide variety and large quantity of data ensuring that usually data trends are statistically significant (due to law of large numbers).

Chapter 5:

1 (15 points)

Refer back to Table 5.2, which presents characteristics of people who utilized telemedicine services. Describe the results shown in the table. Suppose you wanted to conduct a survey of the use of telemedicine in your own community.

Overall the prevalence of telemedicine services is 37% with 42% prevalence in women and 31.7% in men. Note that this data is prevalence in the demographics not the percent of telemedicine users that fall into the demographics. The highest race / ethnicities that use telemedicine were Non Hispanic Whites (39.2) and Native Americans (40.6) while Asians, Blacks, Hispanics all roughly had 33% prevalence. The regions with highest prevalence were Northeast and the West (40 and 42.4) while Midwest and South had a lower prevalence (33.3 and 34.3). Additionally usage seemed to follow the trend of having higher prevalence in metropolitan areas as well as older age groups where the more urbanized / higher aged lead to higher levels of prevalence.

For this I will conduct a survey on the uses of telemedicine based on the urbanization level of the individual.

a) How would you choose the participants?

I would perform a stratified sampling by ensuring relatively equal number of data points for the different urbanization levels.

b) What questionnaire items would you include in the survey?

The questions I would focus on in my survey are: 1. What is your urbanization level? (options same as the table's categories) 2. For appointments do you prefer in person, phone, or video? 3. What is your primary use of telemedicine? (Make Appointments, Take Appointments, Refill Prescriptions, N/A [Don't Use Telemedicine])

c) What type of study design is a survey?

A study design is an example of a cross sectional study because it collects data at one point in time.

2 (15 points)

State three uses for descriptive epidemiologic studies. How could descriptive epidemiologic studies examine the following health issues?

1. Permit evaluations of trends in health and disease
2. Provide a basis for planning, provision, and evaluation of health services
3. Identify problems to be studied by analytical methods and suggest areas that might be fruitful for investigation > > Overall descriptive epidemiologic studies focus on the who, where, and the when.

a) The obesity epidemic in the United States

Descriptive epidemiologic studies might be able to determine which populations have a higher prevalence of obesity (ie urbanization levels, geographic locations, genetic history) as well as analyze any secular trends over the years. For example a descriptive study might find that there has been a steady upward secular trend of obesity since the 80s.

b) Increases in the prevalence of type 2 diabetes among adolescents

Descriptive epidemiologic studies might be able to determine which populations have a higher prevalence of type 2 diabetes (ie urbanization levels, geographic locations, genetic history) as well as analyze any secular trends over the years. For example a descriptive study might find that there has been a steady upward secular trend of type 2 diabetes in children (in this case specifically adolescents).

c) Abuse of opioid drugs

Descriptive epidemiologic studies might be able to determine which populations have a higher prevalence of opioid abuse (ie urbanization levels, geographic locations, genetic history) as well as analyze any secular trends over the years. For example a descriptive study might find that there is an increase of opioid use in urbanized areas versus rural areas.

3 (10 points)

Define the terms case reports and case series. Indicate how they are similar and how they differ. Search the internet for examples of case reports and case series of disease

Case reports refer to a single report of a notable health incident while case series refer to a collection of reports grouped by common features (ie symptoms of affected patients). They are similar as they both are referring to a single disease / health event but they differ from the fact that case reports refer to the individual reports while case series refers to a collection of the reports. Another way of thinking about the difference is a case report is the descriptive report of a single patient while the case series refers to a collection of patients with the same disease.

Example I found on the internet is “A rare case of recurrent pericarditis induced by iodinated contrast media” which is a case report of a single individual while “Outcomes of patients with recurrent pericarditis treated with Colchicine: A Case Series of 20 patients” refers to a case series of 20 patients and their health outcomes.